

1. General Description

The EMS3580 is a Dual SPDT analog switch with ultra-low distortion, high OFF-Isolation for special stereo audio applications with negative swing audio signals capacity that features ultra-low R_{on} of 0.16Ω (typical) at $3.3V V_{CC}$.

The EMS3580 operates a single power supply over a wide range from $3.0V$ to $4.5V$ and $1.8V$ logic compatible with high PSRR. With a special pop-click shunt circuitry at each signal pin that eliminates pops and clicks associated at any application conditions likes switched, enable/disable, power-down and power-up.

With superior THD+N performance and other high performance, the EMS3580 is an ideal device for Hi-Fi system applications.

The EMS3580 is available in QFN1826-16L package. All product is Pb-free and Halogen-free.

2. Features and Benefits

- Single supply range operating from $3.0V$ to $4.5V$
- $-118dB$ THD+N into $100k\Omega$ load at $2V_{rms}$
- $-114dB$ THD+N into 32Ω load at $2V_{rms}$
- Signal-to-Noise (SNR) Ratio $132dBA$
- $100dB$ PSRR at $10kHz$
- $145dB$ crosstalk & separation
- Pop/Click shunt circuit
- Audio Path Soft Turn-On/Off for Pop & Click Elimination

3. Applications

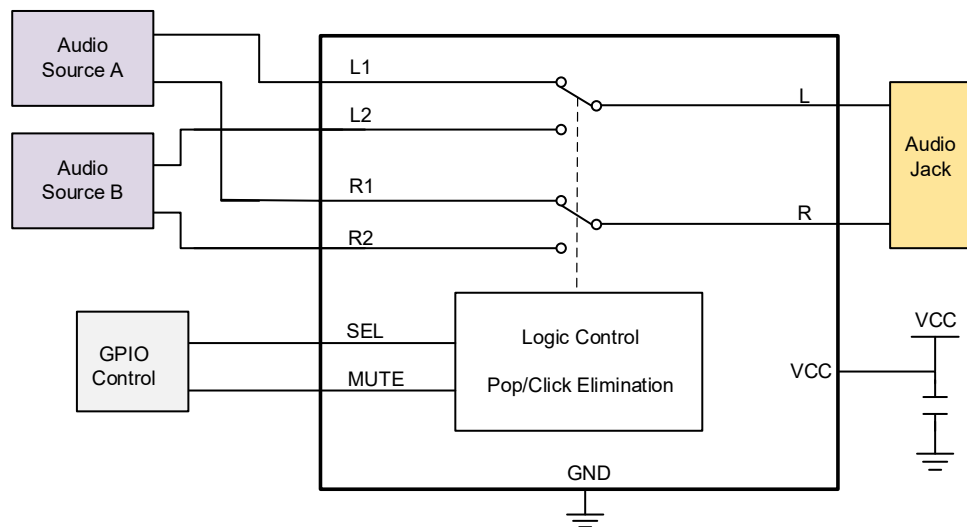
- Hi-Fi Smartphones and Portable Device
- Hi-Fi SACD/DVD Players
- High Quality Home Theaters

4. Ordering Information

Table 1. Ordering information

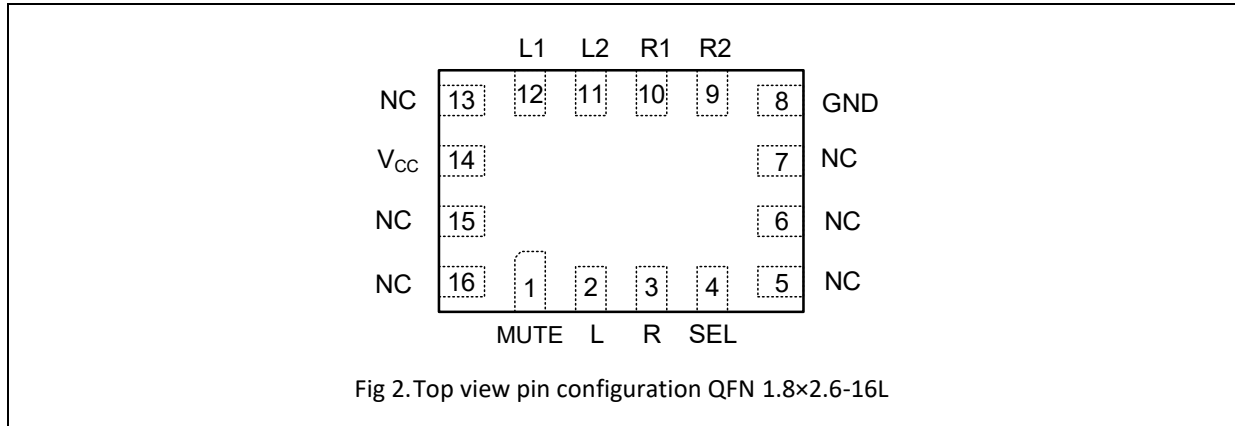
Type number	Package		
	Name	Description	Quantity
EMS3580RSV	QFN1.8×2.6-16L	QFN package, 16 pins 1.8 mm × 2.6 mm; 0.6 mm (Max) height	3000

5. Functional Diagram


Fig 1. Functional diagram

6. Pinning Information

6.1. Pin map



6.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
L, R	2, 3	Left and right channel Common port
R1, L1	10, 12	Left and right channel port 1, normally closed
R2, L2	9, 11	Left and right channel port 2, normally open
MUTE	1	Enable control, active high
SEL	4	Port selection control pin
V _{CC}	14	Power supply
GND	8	Ground
NC	5, 6, 7, 13, 15, 16	Not connection

7. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level; X = Don't care.

Input		Switch	
MUTE	SEL	L1, R1	L2, R2
L	L	ON	OFF
L	H	OFF	ON
H	X	OFF	OFF

8. Absolute Maximum Ratings

Absolute Maximum Ratings may cause permanent damage to the device. This is a stress only rating and operation of the device at these or any other conditions beyond those indicated in the operational sections of this specification is not implied.

Table 4. Absolute Maximum Ratings

Symbol	Parameter	Min	Max	Unit
V _{CC}	supply voltage	-0.3	5.5	V
V _{IN}	digital control input voltage	-0.3	5.5	V
V _{IS}	analog input/output voltage (L1, L2, R1, R2, L, R)	-4.0	4.0	V
I _{IO}	switch continuous current (L1, L2, R1, R2, L, R)		±300	mA
I _{IO_PK}	switch peak current (L1, L2, R1, R2, L, R) (pulsed at 1ms, 10% duty cycle, Max)		±500	mA
P _D	power dissipation in still air		250	mW
T _{STG}	storage temperature range	-55	150	°C
T _J	junction temperature		150	°C
T _L	lead temperature (Soldering, 10 seconds)		260	°C
R _{θJA}	thermal resistance		80	°C/W
I/O to V _{CC} , I/O to GND	ESD protection (HBM)		±6000	V
I/O to I/O			±4000	V

9. Recommended Operating Conditions

Table 5. Recommend operating ratings

Symbol	Parameter	Min	Max	Unit
V _{CC}	supply voltage	3.0	4.5	V
V _{IN}	digital control input voltage	0	V _{CC}	V
V _{IS}	analog input/output voltage (L1, L2, R1, R2, L, R)	-3.3	V _{CC}	V
T _A	operating temperature	-40	85	°C

10. Electrical Characteristics

Table 6. Electrical characteristics

T_a = 25 °C, V_{CC}=3.6V, V_{IS}=2V_{rms}, R_L=32Ω, f=1kHz, unless otherwise noted

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Analog Switch Characteristics						
V _{IS}	analog signal range	V _{DD} = 3.3 V to 4.2 V		2.5		V _{rms}
R _{ON}	on-resistance	V _{IS} = -3.3 V to 3.3V I _{OUT} = 100 mA		0.16		Ω
ΔR _{ON}	R _{ON} matching between channels [1]	V _{IS} = -3.3 V to 3.3V I _{OUT} = 100 mA		0.007		Ω
R _{FLAT(ON)}	R _{ON} flatness [2]	V _{IS} = -3.3 V to 3.3V I _{OUT} = 100 mA		0.001		Ω
Dynamic Characteristics						
THD+N	total harmonic distortion	f = 10 Hz to 22 kHz V _{IS} = 2 V _{rms} @ R _L = 100 kΩ		-118		dB
		f = 10 Hz to 22 kHz V _{IS} = 2 V _{rms} @ R _L = 32 Ω		-114		dB
		f = 10 Hz to 500 kHz V _{IS} = 1.55 V _{rms} @ R _L = 100 kΩ		-104		dB
IMD	intermodulation distortion	Mode = CCIF 19 k + 20 k Ratio = 1 V _{IS} = 500 mV _{rms} @ R _L = 100 kΩ		-122		dB
	dynamic/transient intermodulation distortion	Mode = DIM 100 V _{IS} = 1 V _{rms} @ R _L = 100 kΩ		-103		dB
SNR	signal-to-noise ratio	f = 10 Hz to 22 kHz, inputs grounded R _L = 32 Ω or 100 kΩ		132		dBA
IMB	stereo channel imbalance L1 and R1, L2 and R2	f = 10 Hz to 22 kHz, R _L = 100 kΩ		±0.003		dB
OIRR	off isolation (Muting)	f = 10 Hz to 22 kHz, V _L = V _R = 2 V _{rms} @ R _L = 100 kΩ MUTE = V _{CC} , SEL = "X"		145		dB
		f = 10 Hz to 22 kHz, V _{Lx} = V _{Rx} = 2 V _{rms} @ R _L = 100 kΩ MUTE = V _{CC} , SEL = "X"		145		dB

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High Fidelity Stereo SPDT Switch with Pop and Click Elimination

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
		f = 10 Hz to 22 kHz, V _L = V _R = 2 V _{rms} @ R _L = 32 Ω MUTE = V _{CC} , SEL = "X"		127		dB
		f = 10 Hz to 22 kHz, V _{Lx} = V _{Rx} = 2 V _{rms} @ R _L = 32 Ω MUTE = V _{CC} , SEL = "X"		127		dB
X _{talk}	crosstalk (Channel-to-Channel) [3]	f = 10 Hz to 22 kHz, V _{IS} = 2 V _{rms} , source impedance = 0 Ω R _L = 100 kΩ		145		dB
PSRR	power supply ripple rejection	f = 10 kHz, V _{IS} = 0.1 V _{rms} , inputs grounded		100		dB
BW	-3dB bandwidth	R _L = 50 Ω		50		MHz
T _{TRS-OM}	on-to-mute time			57		ms
T _{TRS-MO}	mute-to-on time			35		ms
T _{OFF}	turn-off time	V _{IS} =1.5V, R _L = 20kΩ MUTE=0		3.5		us
T _{ON}	turn-on time	V _{IS} =1.5V, R _L = 20kΩ MUTE=0		47		ms
T _{BBM}	break-before-make time	V _{IS} =1.5V, R _L = 20kΩ MUTE=0		45		ms
C _{OFF}	Lx, Rx off capacitance	f=100kHz, V _{Lx} or V _{Rx} = V _L or V _R = 0 V		56		pF
C _{ON}	L, R on capacitance	f=100kHz, V _{Lx} or V _{Rx} = V _L or V _R = 0 V		84		pF
Power Supply Characteristics						
I _{CC}	supply quiescent current	MUTE = 0 V		160		μA
		MUTE = V _{CC}		80		μA
Digital Input Characteristics						
V _{IH}	digital input logic high level	V _{CC} = 3.6 V to 4.5 V	1.6			V
		V _{CC} = 3 V to 3.6 V	1.5			V
V _{IL}	digital input logic low level	V _{CC} = 3.6 V to 4.5 V			0.5	V

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High Fidelity Stereo SPDT Switch with Pop and Click Elimination

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
		$V_{CC} = 3\text{ V to } 3.6\text{ V}$			0.4	V
I_{IN}	digital input leakage current				± 2.0	μA
R_{PD}	SEL pull-down resistor			4		$\text{M}\Omega$
R_{PU}	MUTE pull-up resistor			43		$\text{M}\Omega$

[1] R_{ON} matching between channels is calculated by subtracting the channel with the lowest max R_{ON} value from the channel with the highest max R_{ON} value.

[2] Flatness is defined as the difference between maximum and minimum value of ON-resistance at the specified analog signal voltage points.

[3] Crosstalk is inversely proportional to source impedance.

11. Typical Performance Curves

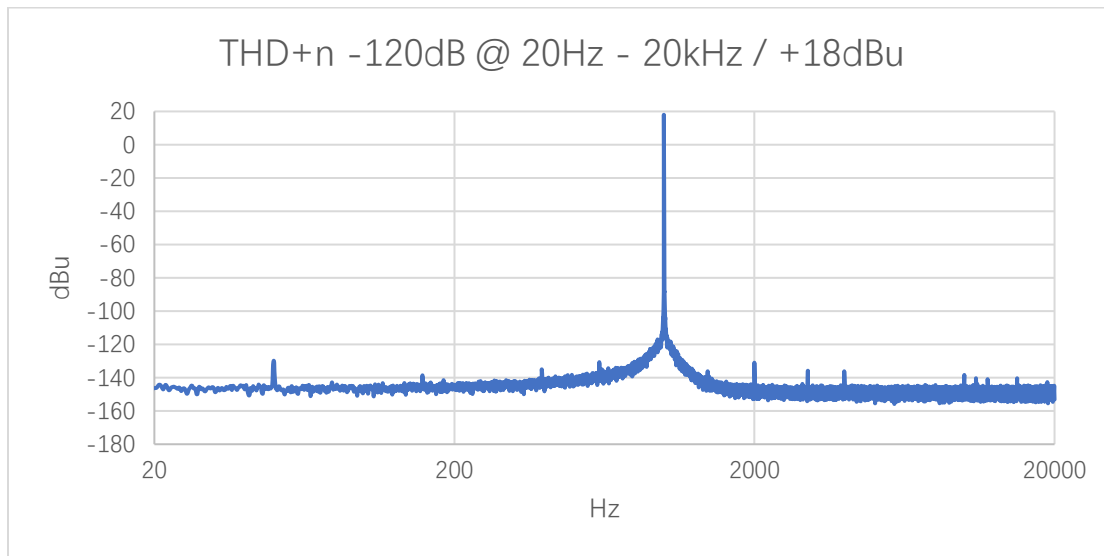


Fig 3. Typical THD+n / 1kHz Tone @ 20Hz - 20kHz (Measurement Limit)

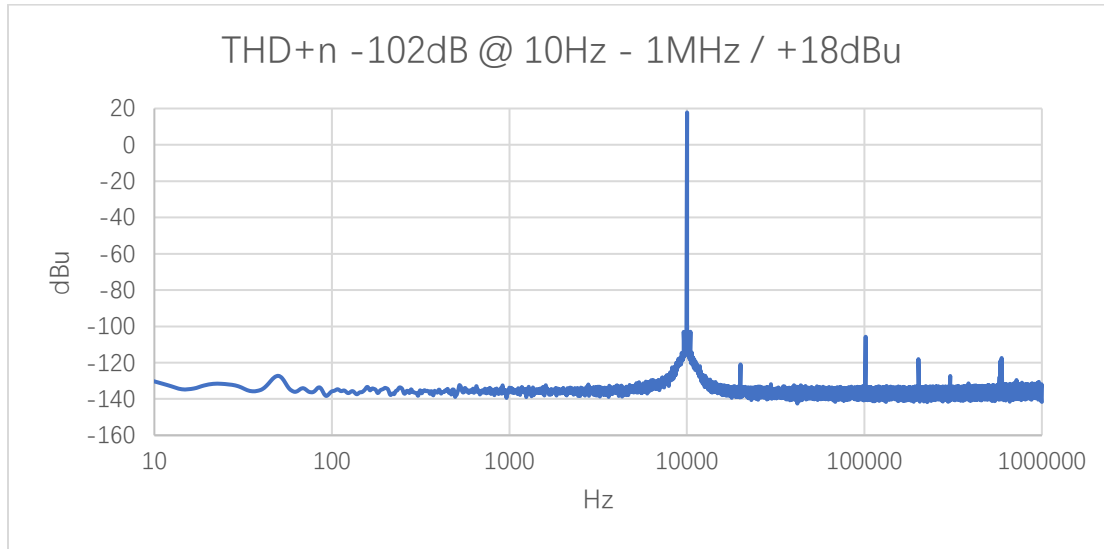


Fig 4. Typical THD+n / 10kHz Tone @ 10Hz - 1MHz (Measurement Limit)

11.1 Source Impedance

Higher source impedance will degrade THD performance, so please design the source circuit carefully, add front buffer circuit for lowest source impedance is good solution generally.

11.2 Signal Swing

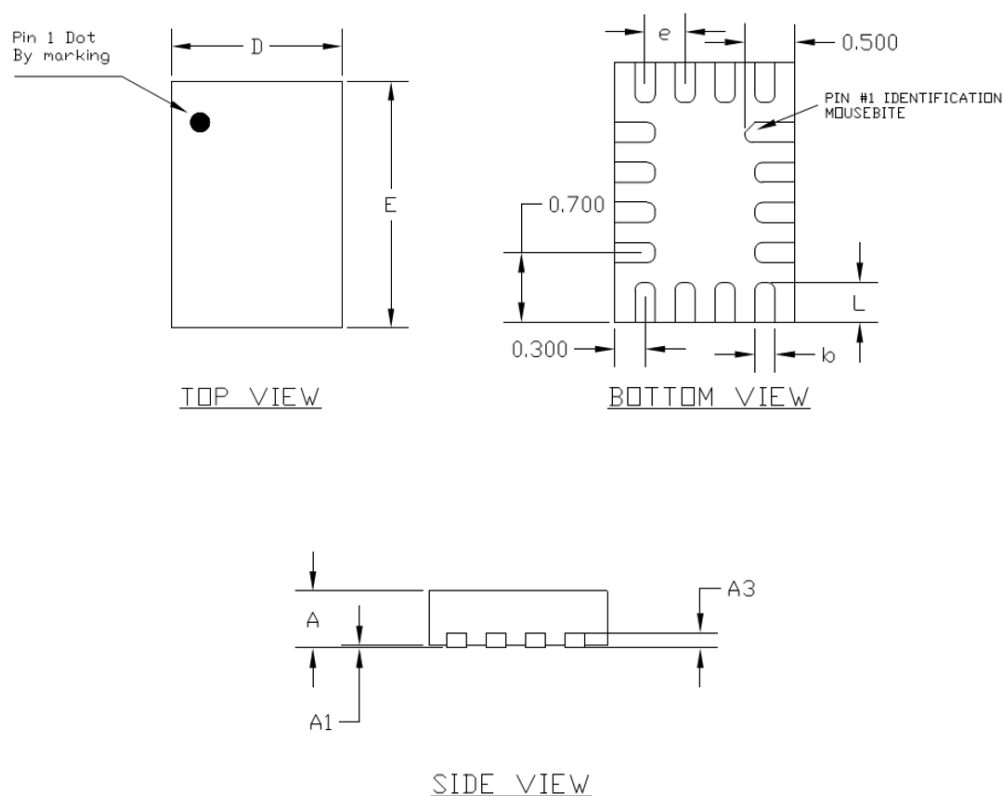
The EMS3580 design for +18 dBu(6.15Vrms) Differential audio system. The damage limit is +20.5dBu(8.2Vrms) Differential, >2dB safe margin is retained.

Table 7. Recommend Signal Swing Amplitude

VCC = 3.3V	VCC = 3.6V	VCC = 3.9V	VCC = 4.2V	VCC = 4.5V
2.3 V _{rms} SE	2.4 V _{rms} SE	2.7 V _{rms} SE	2.9 V _{rms} SE	3.1 V _{rms} SE
4.6 V _{rms} DIFF	4.8 V _{rms} DIFF	5.4 V _{rms} DIFF	5.8 V _{rms} DIFF	6.2 V _{rms} DIFF

12. Package Outline

QFN 1.8x2.6 -16L



Symbol	Dimension in Millimeters		
	Min.	Typ.	Min.
A	0.50	0.55	0.60
A1	0.00	-	0.05
A3	0.15 Typ.		
D	1.75	1.80	1.85
E	2.55	2.60	2.65
L	0.30	0.40	0.50
b	0.15	0.20	0.25
e	0.40 Typ.		

12. Abbreviations

Table 8. Abbreviations

Acronym	Description
ESD	ElectroStatic Discharge
HBM	Human Body Model

13. Revision History

Table 9. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EMS3580 Rev. 1.0	Feb 21, 2026	Product datasheet		